



MYSTRA AUTOMATION

EXCLUSIVE PROPOSAL

Powering the Future of Healthcare in **Enugu**

A Full-Stack AI Automation Proposal for the
Enugu International Hospital — Rangers Avenue, Enugu City

FACILITY
Enugu International Hospital

CAPACITY
300 Beds · Quaternary

PREPARED BY
Mystra Automation

DATE
June 2026

"Where precision meets care." | mystrahq@gmail.com

WHO WE ARE

We build AI that works for Africa.

Mystra Automation is a next-generation AI solutions agency purpose-built for African businesses and institutions. We design, deploy, and manage intelligent automation systems — from conversational AI and voice agents to smart building management, patient data platforms, and real-time analytics dashboards.

9

Automation product categories

AI-First

Every system is intelligence-led

1

Partner for your entire tech stack

Africa

Built for local infrastructure realities

WHAT WE BRING TO THIS PROJECT

End-to-End Ownership

We don't just recommend technology — we design it, build it, integrate it, and maintain it. The hospital deals with one vendor, not 15.

Nigeria-Based Team

Deep local implementation experience. We understand power realities, connectivity constraints, and the Nigerian healthcare workflow.

Hardware + Software

Vendor procurement, hardware sourcing, server setup, IoT deployment, and cloud infrastructure — all under one contract.

24/7 SLA Support

Post-launch monitoring, software updates, staff training, and a dedicated account manager. Your technology never goes unmanaged.

THE CONTEXT

Nigeria loses \$1 billion every year. This hospital changes that.

Nigeria spends an estimated **\$1 billion annually on medical tourism** — patients flying to India, the UK, and Germany for treatments that could be delivered locally. Governor Peter Mbah's answer is the **Enugu International Hospital**: a 300-bed quaternary facility on Rangers Avenue, 85% complete as of April 2026, targeting world-class standards across emergency care, oncology, surgery, critical care, and maternity.



But world-class infrastructure alone is not enough. The hospitals patients trust in London, Dubai, and Mumbai share one thing beyond physical buildings: **intelligent automation running every system** — from the moment a patient walks in to the moment they're discharged, billed, and followed up.

WHY NOW

The Window of Opportunity

The hospital is in its **final construction phase**. This is the optimal moment to integrate AI systems — before staff are hired at scale, before workflows are locked in, before legacy habits form. Research shows retrofitting AI into an operational hospital costs **3x more** than building it in from day one. The time to act is now.

THE AUTOMATION STACK

Nine intelligent systems. One hospital that thinks.

PILLAR 01

Patient Intake & AI Triage

Zero paper. Instant ESI priority scoring at every entry point.

PILLAR 02

Appointment Scheduling AI

Self-booking via WhatsApp + web. 20–30% fewer no-shows.

PILLAR 03

AI Diagnostics Support

AI second-read for radiology. Critical findings flagged instantly.

PILLAR 04

Ward Management & Nurse Calls

Real-time bed dashboard. Smart call routing by clinical priority.

PILLAR 05

Pharmacy Automation

Robotic dispensing. 90% reduction in medication check errors.

PILLAR 06

Billing & Insurance AI

Auto ICD-10/11 coding. 60% fewer claim denials. NHIS ready.

PILLAR 07

AI Security & CCTV

120 AI cameras. Falls, breaches, anomalies detected in <3 seconds.

PILLAR 08

Smart Building BMS/HVAC

Self-regulating facility. 25–35% energy savings. Infection control.

PILLAR 09

Telemedicine Platform

White-label app extending specialist reach to all 260 PHCs.

Each pillar detailed on the following pages · All systems integrated into one unified platform

PILLARS 1 – 3

Patient-Facing Intelligence

1 Patient Intake & AI Triage

Digital self-check-in kiosks at all entry points. An AI triage engine scores every patient's Emergency Severity Index automatically – high-risk cases are flagged *before* they reach the nurse desk, not after.

Technology: 6× 15.6" touch kiosks · Custom ML triage model · Integrated with EHR

75.7% ESI accuracy — PMC peer-reviewed study

40% reduction in manual triage time

2 AI Appointment Scheduling

24/7 self-booking via WhatsApp, web portal, and voice agent. AI predicts and back-fills cancellations in real time. Automated reminders eliminate ghost appointments. Nigeria-specific: WhatsApp-first — meets patients where they already are.

Technology: Mystra Booking Engine · WhatsApp Business API · ElevenLabs voice agent

300–500% net ROI · Industry-verified (Medozai/IntuitionLabs 2025)

20–30% no-show reduction

3 AI Diagnostics Support

AI second-opinion layer integrated into the radiology/PACS workflow. Flags anomalies, prioritises critical reads, generates preliminary reports for radiologist review. Directly supports EIH's flagship oncology focus — early cancer detection requires this speed.

Technology: PACS integration · Aidoc / Google Health CXR AI (FDA-cleared) · Pathology API

10–15% faster critical report generation (IntuitionLabs 2025)

Reduces misread risk for oncology screening

PILLARS 4 – 6

Clinical Operations Automation

4 Ward Management & Intelligent Nurse Call

Real-time bed management dashboard combined with AI-powered nurse call routing. The system triages call priority – clinical emergency vs. comfort request – and routes to the right staff automatically. RTLS tags track all critical equipment.

Technology: IoT bed sensors · RTLS equipment tags · 300× smart nurse call panels · Central command dashboard

- 30% reduction in non-clinical nurse task burden
- Full real-time bed occupancy visibility

5 Pharmacy Automation

Robotic central dispensing unit + ward-level Automated Dispensing Cabinets (ADCs). Drugs are counted, verified, and dispensed with zero human error at the packaging stage. Inventory is managed automatically – no stockouts, no drug waste.

Technology: Omnicell XR2 central robot · 10× Omnicell XT ADC cabinets · Swisslog/BD integration · EHR link

- 90% reduction in pharmacy check errors (Omnicell-verified)
- ROI achieved in 24–36 months

6 Billing & Insurance AI

NLP-based medical coding engine reads clinical notes and auto-generates ICD-10/11 codes. Claims are validated before submission. Built for NHIS compliance and private insurance networks operating in Nigeria. Eliminates revenue leakage from manual coding errors.

Technology: NLP coding engine · NHIS API integration · Automated claims management · Revenue cycle dashboard

- 60% reduction in claim denials
- 40% faster revenue cycle turnaround

PILLARS 7 – 9

Facility & Reach Intelligence

7 AI Security & CCTV Analytics

120 AI-powered 4K IP cameras across the 300-bed facility. The system detects anomalies in real time: unauthorized access, crowd formation, patient falls, and perimeter breaches. Alerts are pushed to the security control room in under 3 seconds. Fall detection reduces patient injury liability.

Technology: Hikvision DeepInMind AI 4K cameras · Axis network cams · 40× HID biometric access control

- 24/7 automated monitoring — no manual screen-watching
- <3 second anomaly alert response

8 Smart Building — BMS / HVAC / Energy

A unified Building Management System integrates HVAC, lighting, fire safety, power, and infection control sensors into one AI-driven platform. Real-time occupancy sensing auto-adjusts airflow. Environmental sensors monitor particulates and VOCs — critical for EIH's oncology wards requiring strict sterile-air protocols.

Technology: Siemens Desigo CC or Honeywell EBI · Smart meters · 500× IoT sensors · Automated lighting zones

- 25–35% energy cost reduction (HealthTech Magazine 2025)
- Real-time infection control compliance

9 Telemedicine Platform

White-labeled telemedicine portal and mobile app enabling video consultations, remote prescriptions, lab result delivery, and post-discharge follow-up — all branded as "Enugu International Hospital." Extends specialist reach to Enugu State's full network of 260 Primary Healthcare Centres.

Technology: React Native app · Daily.co / Vonage video API · EHR integration · WhatsApp prescription delivery

- Reaches 260 PHCs — rural patients without travel
- Reduces avoidable readmissions

ARCHITECTURE

Three-Tier AI Architecture

A unified stack — cloud intelligence, on-premises edge processing, and IoT hardware — all integrated into one platform. Data sovereignty compliant: primary compute hosted on AWS/Azure Nigeria data centre.

TIER 1 — CLOUD AI LAYER

› EHR / EMR (Epic or equivalent)

› AI Scheduling Engine

› AI Triage & Intake Processing

› Billing & Claims AI

› Telemedicine Video Platform

› Analytics & Reporting Dashboard

› Hosted: AWS / Azure Nigeria DC

TIER 2 — ON-PREMISES EDGE

› Dell PowerEdge R750 Servers ×2

› Cisco Catalyst 9300 Network Core

› Wi-Fi 6 (Meraki MR57) — 80 APs

› BMS Controller (Siemens Desigo)

› Pharmacy Robot Controller

› Local AI Inference (low-latency)

TIER 3 — HARDWARE & IOT

› 6× Patient Intake Kiosks

› 300× Bedside Patient Tablets

› 120× Hikvision AI 4K Cameras

› 40× Biometric Access Control

› 500× IoT Sensors (env/bed/HVAC)

› Omnicell Pharmacy Robot + ADCs

Vendor independence: Our architecture is built on open APIs and standards. If a vendor is discontinued or outperformed, components can be swapped without rebuilding the entire system. You are never locked in.

PRIMARY EHR

Epic Systems

42.3% global hospital share

PHARMACY AUTOMATION

Omnicell + Swisslog

2024 BD co-marketing agreement

DOCUMENTATION AI

Microsoft DAX Copilot

Voice-to-notes, Epic-integrated

RETURN ON INVESTMENT

The numbers are clear.
AI pays for itself.

Automation Category	Verified ROI / Impact	Source
Scheduling AI — Net ROI	300–500%	Medozai / IntuitionLabs 2025
No-Show Rate Reduction	20–30%	Industry benchmark 2025
AI Triage ESI Accuracy	75.7%	PMC NCBI peer-reviewed study
Pharmacy Check Error Reduction	90%	Omnicell (manufacturer-verified)
Pharmacy Automation ROI Timeline	24–36 months	PMC EU Economic Analysis 2024
Radiology — Faster Critical Reports	10–15%	IntuitionLabs AI 2025
Smart Building Energy Savings	25–35%	HealthTech Magazine 2025
Billing — Claim Denial Reduction	60%	Healthcare industry benchmark
Revenue Cycle Acceleration	40% faster	Healthcare industry benchmark

Global AI Healthcare Market:

\$39.25B in 2025
Growing to **\$504.17B** by 2032

Investment Horizon:

A fully automated 300-bed hospital typically recovers its AI investment within **2–3 years**, then runs net-positive on labor savings, energy savings & revenue cycle gains indefinitely.

BILL OF MATERIALS

Hardware Requirements

Category	Item / Specification	Qty	Purpose
Servers	Dell PowerEdge R750 (64-core, 512GB RAM, 20TB NVMe)	2	On-prem AI inference + EHR
Network Core	Cisco Catalyst 9300 series managed switches	4	Fiber backbone switching
Wi-Fi	Cisco Meraki MR57 — Wi-Fi 6 (802.11ax) APs	80	Full-facility wireless
Intake Kiosks	15.6" touchscreen self-check-in units	6	OPD, Emergency, dept. check-in
Workstations	Dell OptiPlex / HP EliteDesk (clinical + admin)	80	Ward + admin computing
Bedside Tablets	10" ruggedized Android / iPad (IP-rated)	300	Patient portal + nurse rounds
CCTV Cameras	Hikvision DS-2CD2T87G2 AI 4K IP cameras	120	Security + AI video analytics
Access Control	HID biometric readers (fingerprint + card)	40	All entry points + secure zones
Nurse Call	Smart bed-level call panels (integrated alert)	300	Patient-to-nurse alert system
IoT Sensors	Temp / humidity / occupancy / VOC / CO2 nodes	500	BMS + infection control
Pharmacy Robot	OmniceLL XR2 automated dispensing unit	1	Central robotic dispensing
Pharmacy ADC	OmniceLL XT automated cabinets (ward-level)	10	Ward-level medication dispensing
UPS Systems	APC Smart-UPS 3000VA (redundant power)	20	Critical system power protection
BMS Controller	Siemens Desigo CC building management gateway	1	Unified building management hub

Full detailed BOM with unit costs, lead times, and installation specs available in the Technical Scope Document · Delivered within 2 weeks of engagement start.

PROOF POINTS

AI already works in African healthcare.

These are not future projections — these are operational deployments delivering measurable outcomes across the continent right now.

UCH Ibadan

Nigeria

WhatsApp Telemedicine + AI Scheduling Bot

+35%

Specialist utilization increase

Patients in Oyo, Kwara & Osun reached hospital specialists without traveling. Pilot demonstrates Nigeria's readiness for digital-first healthcare delivery.

Kenyatta National Hospital

Kenya

Google Health AI — Chest X-Ray Diagnostics

TB+

Detection rates improved significantly

AI radiology screening reduced radiologist time per case. Kenya's 2025 National AI Strategy cites this as the blueprint for Africa-wide diagnostic AI rollout.

Aga Khan Hospital

Nairobi, Kenya

Epic EHR + Honeywell BMS Integration

28%

Energy cost reduction

500+ beds. Full digital patient records. East Africa's leading private hospital. This is the operational benchmark EIH should target — and exceed.

Nigeria National Policy Alignment: Nigeria's **2024 National AI Strategy** (Federal Ministry of Communications & Digital Economy) explicitly prioritises AI in healthcare. This means EIH's AI investment aligns with federal policy — and may be eligible for co-funding and regulatory fast-tracking.

DEPLOYMENT ROADMAP

18-Month Implementation Plan



PARTNERSHIP MODEL

What Mystra delivers end-to-end.

WHAT MYSTRA DELIVERS

- 1 Full System Design**
Architecture, vendor selection, integration mapping
- 2 Hardware Procurement**
Vendor negotiation, import, customs clearance, delivery
- 3 Installation + Integration**
On-site installation, configuration, system integration
- 4 Custom Software**
Scheduling AI, telemedicine app, billing engine, dashboards
- 5 Training + Hypercare**
All staff + IT team, 90-day on-site post-launch support
- ∞ Ongoing SLA Support**
24/7 monitoring, updates, dedicated account manager

WHAT THE HOSPITAL PROVIDES

Infrastructure Access

Construction-phase access for cabling, sensor mounting, and server room buildout per Mystra spec sheet.

IT Liaison

One dedicated IT point-of-contact during implementation phases.

Staff Cooperation

Participation in training sessions and user acceptance testing by department heads.

Prime Technology Integrator Model: Mystra is your single point of accountability for the entire technology stack. One contract. One team. One SLA. No vendor coordination overhead on your side.

INVESTMENT

Budget Tiers — Choose Your Level

Tier 1 — Core AI Foundation

₹180M – ₹250M

Estimated investment · Hardware + software + integration

- ✓ AI Patient Intake + Triage System
- ✓ Appointment Scheduling AI (WhatsApp + web)
- ✓ Basic Billing + NHIS Integration
- ✓ AI Security CCTV (120 cameras)
- ✓ Smart Building BMS (HVAC + energy)
- ✓ Servers, network core, Wi-Fi 6, kiosks
- ✓ AI Diagnostics (Radiology/Pathology)
- ✓ Pharmacy Automation Robot
- ✓ Telemedicine Platform
- ✓ Bedside tablets (300 units)

Payback horizon: 18–24 months

Tier 2 — Full Intelligence Stack

RECOMMENDED

₹380M – ₹520M

Complete AI ecosystem · Everything Tier 1 includes, plus:

- ✓ Everything in Tier 1
- ✓ AI Diagnostics Support (radiology/pathology)
- ✓ Ward Management + RTLS + Nurse Call
- ✓ Pharmacy Automation (robot + 10 ADC cabinets)
- ✓ Advanced Billing AI + Insurance Automation
- ✓ Telemedicine Platform (mobile app + web)
- ✓ 300× Bedside Patient Tablets
- ✓ Full staff training program
- ✓ 90-day on-site hypercare support

Payback horizon: 24–36 months · Annual savings: ₹80–120M/year post-payback

ANNUAL SLA SUPPORT

₹18M – ₹30M

Per year · Post-launch

INCLUDES

- Software updates
- 24/7 monitoring
- Helpdesk + account manager

PAYMENT STRUCTURE

- 30% mobilisation
- 40% on Phase 2 start
- 30% on go-live

LET'S BUILD THIS

Let's build the smartest hospital in West Africa.

The Enugu International Hospital is already making history. The question is not *whether* to adopt AI — it's whether to do it right from day one, or pay three times more to retrofit it later. Our first deliverable, at **no cost**, is a detailed technical scope and hardware BOM specific to EIH's floor plan and department layout.

- 1

Discovery Meeting

1–2 hours. Facility walkthrough, department head briefing, IT infrastructure assessment. We map your exact needs before proposing a single naira of spend.
- 2

Technical Scope Delivery

Within 2 weeks of the discovery meeting: detailed specification document, vendor shortlist, refined budget with line-item pricing, and integration architecture diagram.
- 3

Contract & Kick-Off

Phase 1 infrastructure begins within 30 days of contract signing. Your hospital opens as one of the most technologically advanced facilities in West Africa.

CONTACT

Mystra Automation

mystrahq@gmail.com

Africa's AI Solutions Agency

OUR COMMITMENT

"Africa deserves world-class healthcare technology.
We're here to build it."

AI-First

Africa-Built

June 2026